

Tofa Kimani Mwangi

Data Engineer | Software Engineering & Big Data

✉ mwangitofa@gmail.com ☎ 0759799624 📍 Nairobi 🌐 <https://tofa-mwangi.vercel.app/>

Professional Summary

Applied Computing graduate and ALX-certified Data Engineer with hands-on experience designing end-to-end ETL pipelines, star-schema data warehouses, and automated analytics workflows. Proficient in Python, SQL, PySpark, Apache Airflow, PostgreSQL, and containerization using Docker. Proven track record of optimizing data integrity and processing large-scale telecom and healthcare datasets for operational reporting

Technical Skills

Programming Language
Python, Javascript, SQL, Java
Data Engineering & Big Data
PySpark, Apache Airflow, ETL/ELT Pipelines, Data Warehousing, Dimensional Modelling(Star Schema), Relational Modeling(RDBMS), Pandas, NumPy
Web & Databases
PostgreSQL, MySQL, Express, Spring Boot, Next.js, Restful APIs, Tailwind CSS, Bootstrap
Tools, Cloud & DevOps
Docker, Git, Github, Jupyter Notebook, Anaconda, Advanced Excel, CI/CD Basics
Methodologies & Core Competencies
Agile/Scrum, Test-Driven Development(TDD), System Optimization, Database Architecture, Prompt Engineering, Technical Troubleshooting

Work Experience

Minerva Analytics	Remote
Corporate Governance Data Analyst - Virtual Assistant	January 2026 – April 2026
- Audited public company filings, annual reports, and proxy statements to log board structure changes, executive leadership shifts, and committee appointments. - Captured and verified financial metrics, director independence data, and board governance updates with high accuracy across global markets.	
Social Health Authority	Nanyuki
IT Intern	September 2024 - December 2024
- Cleaned and prepared large-scale datasets using Excel and internal diagnostic tools, reducing reporting errors and optimizing data integrity. - Conducted operational data analysis to identify bottlenecks and collaborated with cross-functional teams to redesign IT workflows, accelerating report generation. - Resolved technical hardware, software, and network issues across computer systems to ensure high uptime for critical user applications.	

Projects

Zuri Mobile Customer Churn ETL Pipeline PySpark, Apache Airflow, PostgreSQL, Docker Compose	June 2026
- Engineered a production-style telecom churn pipeline processing 10,000 records using PySpark for data validation, deduplication, and standardization. - Designed a PostgreSQL star-schema data warehouse to optimize dimensional modeling and analytics performance. - Orchestrated automated, repeatable staging-to-live data workflows using Apache Airflow and Docker Compose	
https://github.com/TofaKM/zuri-mobile-churn-pipeline	
Hospital Data ETL Pipeline Docker, Python, SQL, ETL Architecture	June 2026
- Developed a containerized ETL pipeline in Docker to automate healthcare data ingestion, cleaning, and transformation. - Structured end-to-end data workflows generating high-precision datasets for operational analysis and clinical reporting. - Automated routine transformations to allow non-technical stakeholders direct access to accurate business insights	
https://github.com/TofaKM/etl_hospital_pipeline	
Chap Mall E-Commerce Platform Next.js, Node.js, PostgreSQL, RESTful APIs	July 2025
- Architected a multi-tier e-commerce platform featuring Role-Based Access Control (RBAC) across customer, vendor, and admin roles. - Engineered RESTful APIs and optimized PostgreSQL database queries to handle complex relational data with high data integrity. - Built inventory tracking logic for live availability updates and implemented secure authentication to lower page latency.	
https://github.com/TofaKM/Advanced-E-commerce-Website-	

Certifications

Data Engineer Certificate	August 2026
ALX Africa	
Key Competencies: ETL/ELT Pipelines, Data Warehousing, Apache Airflow, Apache Spark, Dimensional Modeling, SQL/Python Automation.	
https://ehub.alxafrica.com/ob3/verify-certificate/aac32928-744d-42d4-beff-85b393b73d1a	
Python for Data Science	April 2026
ALX Africa	
Key Competencies: Pandas, NumPy, SciPy, Automated Data Extraction, Statistical Hypothesis Testing	
https://savanna.alxafrica.com/certificates/n8CmfHBhPR	
Data Analytics Certificate	September 2025
ALX Africa	
Key Competencies: MySQL Querying, Power BI Dashboarding, Data Visualization, Professional Reporting, Excel	
https://savanna.alxafrica.com/certificates/97rGSZN3hJ	
Introduction to Machine Learning	February 2025
Cisco IBM	
Key Competencies: Supervised & Unsupervised Learning, Model Evaluation, Predictive Analytics (Python)	
https://www.credly.com/badges/d380bf99-8288-47f9-9883-c848a62d8460/public_url	
Data Analytics	August 2024
Cisco Academy	
Key Competencies: Data Lifecycle Management, Data Cleaning, Advanced Excel, SQL Analysis	
https://www.credly.com/badges/aa459c6a-f252-45e4-b56a-49f9cd516354/public_url	

Education

KCA University	Bachelor of Science in Applied Computing
Computer Science	Nairobi, Kenya • September 2020 - November 2025
Core Coursework: Data Structures & Algorithms (DSA), Systems Analysis & Design, Database Management Systems (DBMS), Relational Database Normalization, SDLC. Capstone Project: Engineered a full-stack e-commerce application using JavaScript, Node.js, React, and MySQL featuring optimized relational schemas and Role-Based Access Control (RBAC).	
Ndururumo High School	High School Certificate
Nyahururu, Kenya • January 2016 - November 2019	
- Achieved Mean Grade B+ (71 points), focusing on Mathematics, Physics, and Computer Studies. - Served as Academic Secretary, leading peer tutoring initiatives and managing academic schedules	

Interests

Professional Interests & Research	Activities & Hobbies
System Architecture, AI & ML, Tech News, Cloud Data Architecture, Real-Time Data Streaming, Distributed System & Microservices	Football, Badminton, Outdoor Activities, Gardening, Animations

Profiles

Portfolio	LinkedIn	GitHub
Tofa Mwangi	Tofa Mwangi	Tofa_KM

References

Dacosta Otekoh Human Resources (HR)	Francis Ngugi Software Engineer
Phone No. - 0701563588	Phone No. - 0707 574430
https://email:dacostaotekoh@gmail.com	https://francisngugi518@gmail.com
Faith Muna Claim Officer / Supervisor Social Health Authority	
Phone No. - 0722613166	